SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26187
- **Problem Statement Title -** City-Wide AI Engine for Multi-Camera ANPR Trajectory Tracking and Urban Traffic Analytics
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 191353
- **Team Name (Registered on portal) – TechNova_2.O**

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IDEA TITLE

CITY-WIDE AI ENGINE FOR MULTI-CAMERA ANPR

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Connecting isolated CCTV/ANPR feeds into a unified vehicle intelligence and traffic analytics platform.

REAL-WORLD ISSUE

Modern cities operate large networks of CCTV and ANPR cameras, but many systems process feeds independently. This makes it difficult to connect vehicle sightings across different locations and reconstruct complete vehicle movement patterns.

WHY IT MATTERS

Without cross-camera tracking and analytics, traffic management, law enforcement and urban planning remain reactive instead of proactive. This leads to congestion, delayed investigations and reduced city safety.

OUR SOLUTION

A centralized AI platform that processes multi-camera feeds to detect vehicles, read license plates, track vehicles across camera locations, reconstruct their trajectories and generate city-wide traffic analytics.

CORE AI ENGINE

Unified Video Analytics Care

HIGH-ACCURACY ANPR

- License plate detection
- OCR
- Challenging conditions
- >90% accuracy (target)

TRAJECTORY TRACKING

- Vehicle/plate ID
- Cross-camera matching
- Movement history
- Direction & route
- GIS visualization

TRAFFIC ANALYTICS

- Traffic density
- OD patterns
- Congestion detection
- Traffic flow trends
- Real-time heatmaps

PERIMETER INTELLIGENCE

Multi-camera correlation
Vehicle re-identification
GIS mapping
Real-time analytics

RESPONSE & COMMAND

Live dashboard • Alerts • Event logs • Evidence



RISK ▶ SOLUTION

Fragmented Camera Feeds
Multiple independent CCTV/ANPR systems

Unified AI Platform
Centralized processing & analytics

No Cross-Camera Tracking
Vehicle re-identification & trajectory engine

Limited Traffic Insights
Real-time analytics & GIS visualization

Delayed Incident Response
Instant alerts & command dashboard

Smarter, Safer Cities
Data-driven traffic management and law enforcement

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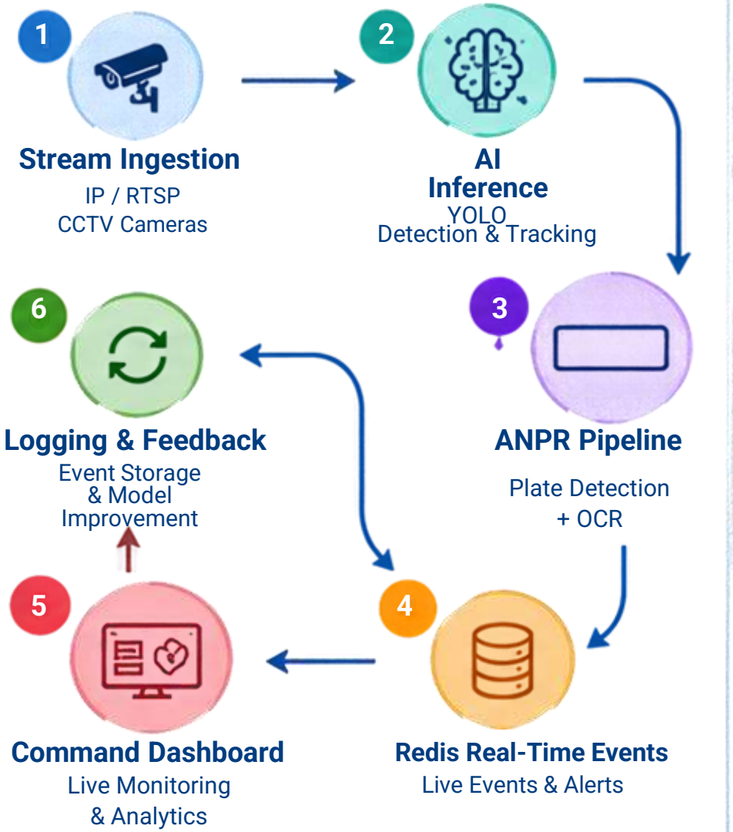
TECHNICAL APPROACH

CITY-WIDE AI ENGINE FOR MULTI-CAMERA ANPR

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Connecting isolated CCTV/ANPR feeds into a unified vehicle intelligence and traffic analytics platform.

PROCESSING METHODOLOGY



SYSTEM ARCHITECTURE

Multi-Camera CCTV/ ANPR Feeds
IP / RTSP Streams (City-wide Network)

Stream Ingestion Layer
FastAPI + WebSocket

Vehicle Detection
YOLO (V8X11) + ByteTrack/BoT-SORT

Plate Detection & Image Enhancement
Image Processing + OCR (ANPR)

Vehicle Tracking + Re-ID
Cross-Camera Matching

Trajectory Reconstruction
Movement History + Direction + Route

Traffic Analytics
Flow, Density, Congestion, Origin-Destination

GIS Command Dashboard
React + Leaflet (Live Map & Analytics)

TECHNOLOGIES USED

AI / Computer Vision
Python OpenCV YOLO

Backend
FastAPI WebSocket REST API

Database
PostgreSQL/MySQL Redis

Frontend
React Tailwind CSS JS HTML
Leaflet

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FEASIBILITY AND VIABILITY

CITY-WIDE AI ENGINE FOR MULTI-CAMERA ANPR

Smarter Traffic | Safer Cities | Data-Driven Decisions

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FEASIBILITY ANALYSIS

Existing Infrastructure Reusability

Leverages existing city CCTV and ANPR camera networks. No need for new or replacement hardware.

Scalable Deployment

Can be deployed across intersections, zones and city sectors in phases.

Modular AI Pipeline

Detection, ANPR, tracking, trajectory and analytics modules can be developed and deployed independently.

Cross-Camera Intelligence

Correlates camera-level observations to reconstruct vehicle movement across the network.

24/7 Automated Monitoring

AI-driven system ensures continuous surveillance without fatigue, improving efficiency and reducing manpower dependency.

VIABILITY

Proven Concept

Built on proven AI/Computer Vision and ANPR technologies already successful in surveillance and traffic analytics worldwide.

Builds Operational Trust

Automated detection minimizes human bias and provides consistent, reliable monitoring across multiple locations.

Ensures Accountability

All detections are timestamps, logged and traceable with video evidence for transparency and audibility.

Authority Engagement

Real-time dashboards, alerts and event logs keep operators engaged and enable quick decision-making.

Future Ready

Modular architecture allows integration of advanced analytics, machine learning models and new use cases as technology evolves.

BUSINESS POTENTIAL

Enhanced Urban Mobility & Safety

Reduces congestion, enables faster incident response and improves road safety.

Operational Cost Savings

Automates manual monitoring and reporting, reducing manpower costs and optimizing resource allocation.

Faster Response Time

Instant detection and alerts help authorities respond quickly to incidents, violations and emergencies.

Data-Driven Insights

Provides traffic trends, originestination patterns, heatmaps and predictive analytics for better planning and decision-making.

Long-Term Impact

Creates a sustainable, smart and connected transport ecosystem, supporting safer, smarter and more efficient cities.

Feasible to implement with existing resources | **Viable for long-term adoption** | **High impact on urban traffic and national economy**

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IMPACT AND BENEFITS

Smarter Traffic. Safer Cities. Data-Driven Decisions.

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City-Wide Vehicle Visibility

Connects multiple CCTV/ANPR cameras to provide a unified view of vehicle movement.

Faster Vehicle Search

Search a license plate and retrieve its detected journey across connected camera locations.

Traffic Flow Intelligence

Generates traffic density, movement and congestion insights from multi-camera observations.

Data-Driven Urban Planning

Provides origin-destination patterns, heatmaps and analytics for better traffic planning.

Auditable Vehicle Movement

Time stamped detections and trajectory history provide traceable evidence for analysis.

Scalable Smart-City Deployment

Modular software architecture can expand from pilot zones to city-wide camera networks.

IMPACT AT A GLANCE

Connected Camera Intelligence
Unifies multiple CCTV/ANPR feeds for comprehensive monitoring.

Efficient Traffic Operations
Reduces congestion and improves resource utilization.

Real-Time Analytics
Delivers actionable insights for data-driven decisions.

Better Incident Response
Enables quick detection and faster response to unusual activity.

Smarter Urban Mobility
Supports sustainable and efficient city transport systems.



From isolated
City-Wide Veh

ANPR
Real-Time ANPR

Cross-Camera Tracking



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RESEARCH AND REFERENCES

SMARTER ANALYTICS . SAFER ROADS . SMARTER CITIES

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RESEARCH / TECHNICAL BASIS

- Computer Vision for real-time traffic video analytics
- YOLO-based vehicle detection
- Multi-object vehicle tracking using ByteTrack / BoT-SORT
- OCR-based ANPR and license-plate recognition
- Multi-camera vehicle re-identification / cross-camera matching
- Spatial-temporal trajectory reconstruction
- Traffic-flow and congestion analytics
- Real-time alerting and event logging

KEY REFERENCES

- YOLO – real-time vehicle detection
- OpenCV – computer vision and video processing
- ByteTrack / BoT-SORT – multi-object tracking
- OCR / ANPR research – license plate recognition
- Vehicle Re-ID – cross-camera matching
- Spatial-temporal trajectory analysis
- FastAPI & WebSocket – real-time communication
- Redis – real-time event processing
- PostgreSQL – vehicle/event data storage
- GIS / Leaflet – route and traffic visualization

SYSTEM ARCHITECTURE

- **CCTV CAMERAS** — Existing road-side camera infrastructure
- **VIDEO STREAM** — Live video input
- **AI DETECTION** — YOLO + OpenCV
- **OBJECT TRACKING** — Multi-object tracking
- **ANPR / OCR** — License plate recognition
- **TRAJECTORY ANALYSIS** — Path and movement analysis
- **EVENT & TRAFFIC DATA** — Data aggregation and storage
- **REAL-TIME DASHBOARD** — Visualization & monitoring

"From Isolated Cameras to City-Wide Vehicle Intelligence"

